AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Spring 2025-2026

Section: [EE]

Group No: 00

Project Report On

Food Ordering CRUD Application

Supervised By

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Submitted By:

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Obtained Marks for CO2 and CO3 (Description given in the following page)

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)

Evaluation Criteria (CO2)	Total =	Evaluation Criteria (CO3)	Total =
Requirement fulfillment		Organization of the application	
Validation		Representation and Integration of Database	
Verification		Graphical User Interface	

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	**Not Attended/ Incorrect (0)**	**Inadequate (1-2)**	**Average (3)**	**Good (4)**	**Excellent (5)**
Evaluation Criteria	**Evaluation Definition**				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or implementation	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive

	of verification processes, and data flow is not considered.	inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	ensuring data integrity and appropriate data flow within the system.	and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

CO3: Prepare and Explain a real-life desktop-based application synthesizing several components of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself.	Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	Exhibits an exceptional understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent.	Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	Effectively identifies and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system.	Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way according to the coherent usage .

1. Title

Food Ordering CRUD Application

2. Introduction

In today's fast-paced digital lifestyle, the demand for quick and convenient food ordering has grown significantly. Ensuring a smooth and efficient ordering process is essential for enhancing customer satisfaction and maintaining the success of food service platforms. This case study explores how a food ordering application operates to provide users with a seamless experience from browsing menus to receiving their meals. It focuses on the interaction between customers, restaurants, and the system itself, highlighting how each component works together to ensure accurate orders, timely preparation, and reliable delivery.

3. Case Study

In a comprehensive food ordering management system, users are empowered to browse and select meals based on a wide range of criteria, allowing them to tailor their dining experience according to their tastes and preferences. Additionally, users can customize their selected orders by modifying ingredients, portion sizes, or adding special instructions. Each user in the system is associated with essential information such as their name, user ID, email, phone number, delivery address, and account credentials.

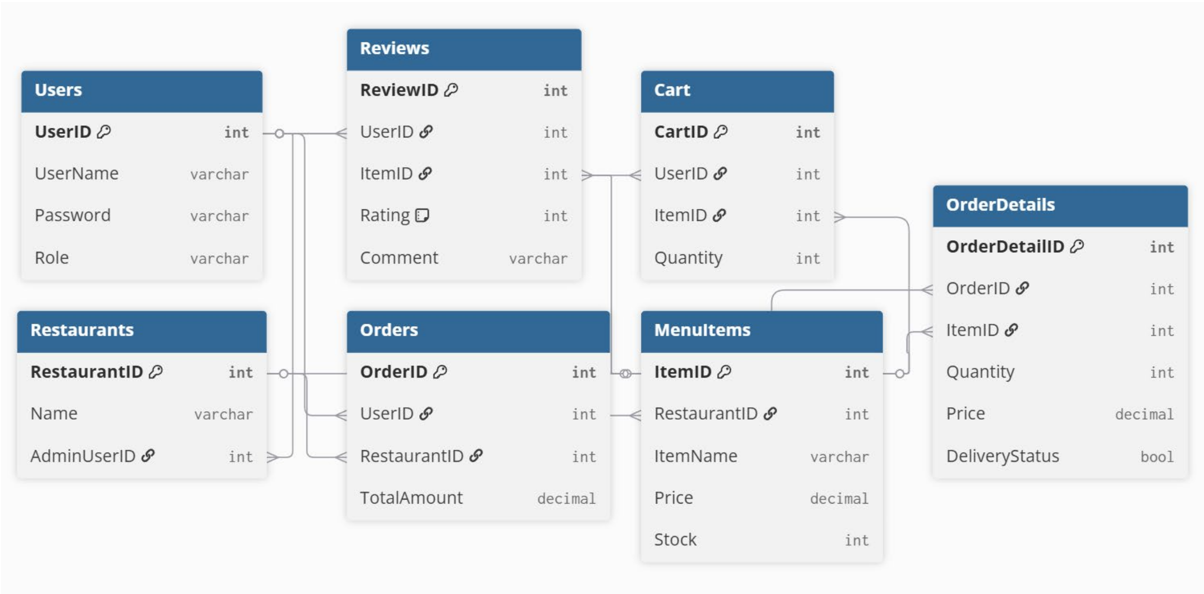
Every food item within the system is uniquely identified by its ID and includes detailed attributes such as the item name, category, price, description, and availability status. Food items are organized and offered by multiple restaurants registered on the platform. Each restaurant can create, update, and manage a variety of food items, but every food item is exclusively associated with a single restaurant.

Restaurants are defined by attributes including their restaurant ID, name, location, contact number, and operational hours. They are responsible for maintaining their menus and ensuring accurate pricing and availability of items. Restaurants can also receive and process orders placed by users, updating order statuses accordingly.

The system facilitates full CRUD (Create, Read, Update, Delete) operations, enabling users to place new orders, view their order history, modify existing orders before confirmation, and cancel orders when necessary. Each order is uniquely identified and contains details such as order ID, list of selected items, total cost, order status, and timestamp.

The platform operates under the supervision of an administrator who holds a unique ID and password. The administrator ensures system integrity, manages users and restaurants, monitors transactions, and enforces security measures across the application.

4. Database Schema



4. Database Query

User Table

```
CREATE TABLE Users (
    UserID INT IDENTITY(1,1) PRIMARY KEY,
    UserName VARCHAR(50) UNIQUE NOT NULL,
    Password VARCHAR(255) NOT NULL,
    Role VARCHAR(20) NOT NULL CHECK (Role IN ('SuperAdmin', 'Admin', 'Customer'))
);
```

Restaurants Table

```
CREATE TABLE Restaurants (
    RestaurantID INT IDENTITY(1,1) PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    AdminUserID INT,
    FOREIGN KEY (AdminUserID) REFERENCES Users(UserID)
);
```

MenuItems Table

```
CREATE TABLE MenuItems (
    ItemID INT IDENTITY(1,1) PRIMARY KEY,
    RestaurantID INT NOT NULL,
    ItemName VARCHAR(100) NOT NULL,
    Price DECIMAL(10,2) NOT NULL,
    Stock INT NOT NULL DEFAULT 0,
    FOREIGN KEY (RestaurantID) REFERENCES Restaurants(RestaurantID)
);
```

Orders Table

```
CREATE TABLE Orders (
   OrderID INT IDENTITY(1,1) PRIMARY KEY,
   UserID INT NOT NULL,
   RestaurantID INT NOT NULL,
   TotalAmount DECIMAL(10,2) NOT NULL,
   FOREIGN KEY (UserID) REFERENCES Users(UserID),
   FOREIGN KEY (RestaurantID) REFERENCES Restaurants(RestaurantID)
);
```

OrderDetails Table

```
CREATE TABLE OrderDetails (
   OrderDetailID INT IDENTITY(1,1) PRIMARY KEY,
   OrderID INT NOT NULL,
   ItemID INT NOT NULL,
   Quantity INT NOT NULL,
   Price DECIMAL(10,2) NOT NULL,
   DeliveryStatus BIT DEFAULT 0, -- 0 = Not Delivered, 1 = Delivered
   FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
   FOREIGN KEY (ItemID) REFERENCES MenuItems(ItemID)
);
```

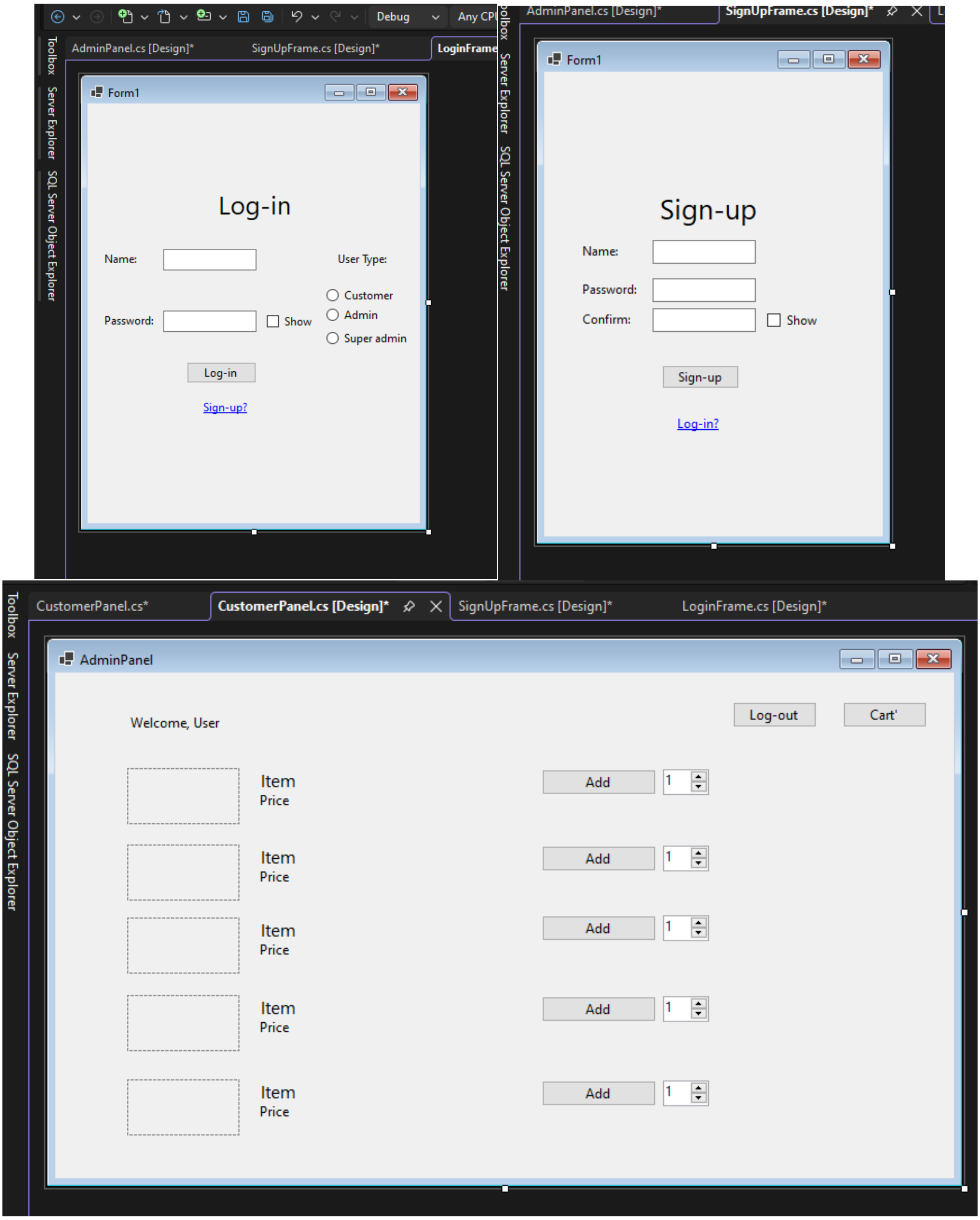
Cart Table

```
CREATE TABLE Cart (
   CartID INT IDENTITY(1,1) PRIMARY KEY,
   UserID INT NOT NULL,
   ItemID INT NOT NULL,
   Quantity INT NOT NULL,
   FOREIGN KEY (UserID) REFERENCES Users(UserID),
   FOREIGN KEY (ItemID) REFERENCES MenuItems(ItemID)
);
```

Reviews Table

```
CREATE TABLE Reviews (
   ReviewID INT IDENTITY(1,1) PRIMARY KEY,
   UserID INT NOT NULL,
   ItemID INT NOT NULL,
   Rating INT CHECK (Rating BETWEEN 1 AND 5),
   Comment VARCHAR(255),
   FOREIGN KEY (UserID) REFERENCES Users(UserID),
   FOREIGN KEY (ItemID) REFERENCES MenuItems(ItemID)
);
```

5. GUI Design



Note: None of the GUI Designs are final and are subject to change